

Semester Project (Sports Car Showroom Management System)



Submitted by:

Wajeeha Javed	2024-CS-601
Minahil Khalid	2024-CS-604
Rohma Shahid	2024-CS-631

Submitted to:

Mr. Zeeshan Ramzan

**Department of Computer Science
University of Engineering and Technology New
Campus**

Proposed Methodology/System:

The development of the Digital Sports Car Management System (DLMS) will follow the Waterfall Model. This methodology is selected for its structured, sequential approach, with each phase completed fully before moving to the next.

The key reasons for choosing the Waterfall Model include:

- Clear Structure: Each phase has defined deliverables and a review process.
- Documentation-Driven: Comprehensive documentation is produced at each stage.
- Predictability: Timeline, cost, and scope are determined early in the project.
- Ease of Management: Progress is easily measurable against phase completion.
- Well-Suited for Stable Requirements: Works effectively when requirements are clearly understood upfront.

Phases in the Project:

1. Requirements Gathering & Analysis

- All functional and non-functional requirements are collected and documented.
- Requirement specification is reviewed and approved before proceeding.

2. System Design

- Architectural design, database design, and interface layouts are created.
- Design documentation is finalized and signed off.

3. Implementation

- The system is developed according to the design specifications.
- Code is written in modules, with unit testing conducted by developers.

4. Testing

- System testing, integration testing, and user acceptance testing are performed.
- Bugs are fixed, and the system is validated against requirements.

5. Deployment

- The system is installed in the production environment.
- Data migration and user training are conducted.

6. Maintenance

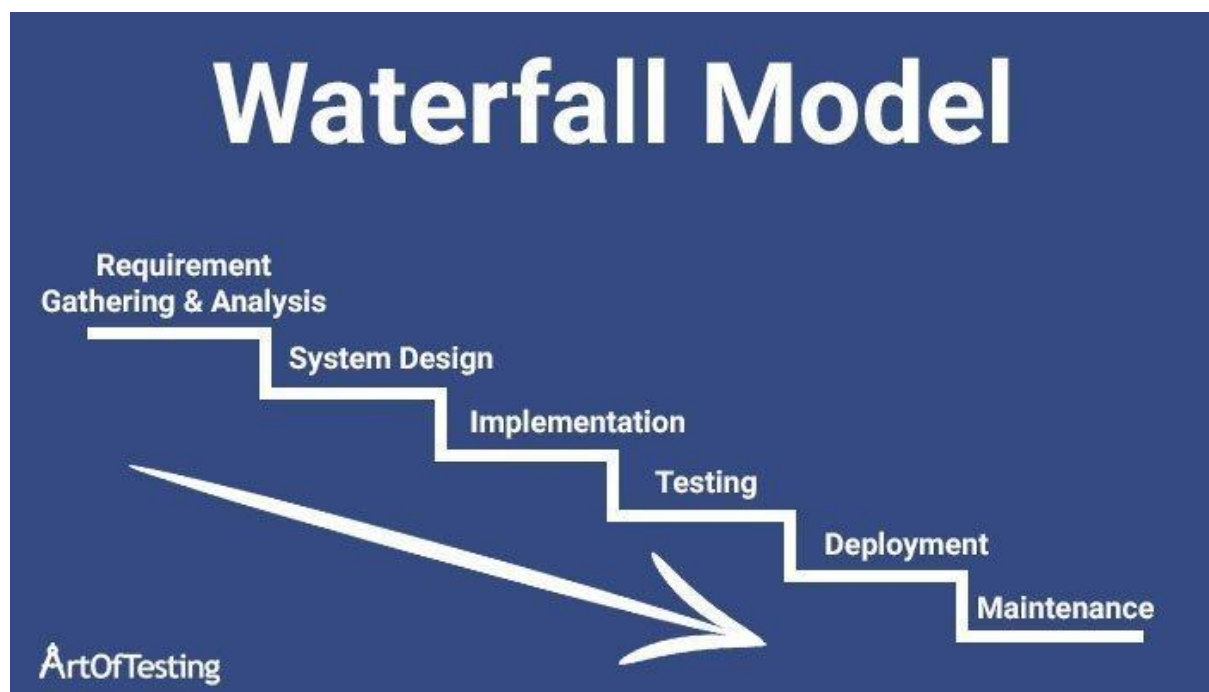
- Ongoing support, updates, and troubleshooting after go-live.

Roles in the Project:

- **Project Manager:** Sir Zeeshan
- **Lead Developer:** Minahil
- **Development Team:** Rohma, Wajeeha
- **QA Testers:** Assigned during the testing phase

Waterfall Process Overview:

Each phase must be completed and approved before the next begins. Changes after a phase is completed require formal change requests and may impact timeline and cost.



Related Work:

Existing System Analysis

When examining existing digital Sports Car Management Systems, several prominent solutions emerge as key players in the automotive retail industry. While these systems offer powerful features and proven reliability, they are primarily designed for large-scale dealerships and often fail to align with the specific requirements of small to medium-sized luxury sports car showrooms. This section analyzes three widely used systems and evaluates their suitability in comparison to the system proposed in this project.

CDK Global [1]

CDK Global is one of the most well-established Dealership Management Systems (DMS) in the automotive industry. It is widely adopted by large franchise dealerships due to its extensive functionality, which includes inventory management, service operations, financing, and accounting.

Strengths:

CDK Global's greatest strength lies in its robustness and scalability. It can efficiently manage thousands of vehicles across multiple departments and locations. The system benefits from strong vendor support, frequent updates, and seamless integration with major automobile manufacturers. Its all-in-one architecture makes it particularly effective for large corporate dealerships with complex operational requirements.

Weaknesses:

Despite its capabilities, CDK Global presents significant drawbacks for smaller showrooms. The system is expensive to deploy and maintain, often requiring long-term contracts and specialized IT support. Its extensive feature set can feel

overwhelming for boutique dealerships, where many tools remain unused. Additionally, its legacy architecture has led to reports of slow performance and a dated, unintuitive user interface, which can hinder sales staff who require speed and simplicity during customer interactions.

DealerSocket [2]

DealerSocket is another widely recognized automotive management system, best known for its strong Customer Relationship Management (CRM) capabilities and support for multi-dealership communication.

Strengths:

DealerSocket excels in managing customer relationships and marketing data, making it well-suited for dealership networks that emphasize customer retention and coordinated sales efforts. Its advanced reporting and data customization features allow managers to track detailed performance metrics and customer behavior.

Weaknesses:

However, DealerSocket has a steep learning curve and often requires extensive staff training. For smaller showrooms, the system's complexity can be counterproductive, as many features are unnecessary for day-to-day operations. Over-customization may result in a cluttered interface, making navigation difficult for new or non-technical users and potentially slowing down sales processes.

AutoManager (WebManager) [3]

AutoManager is a cloud-based inventory management system designed specifically for small, independent car dealerships. It focuses on core functionality such as vehicle listings, website integration, and regulatory compliance.

Strengths:

AutoManager is highly user-friendly and requires minimal training, making it ideal for smaller businesses with limited technical resources. Its cloud-based infrastructure allows for rapid deployment without the need for expensive hardware or on-site servers. Additionally, it is cost-effective and well-suited for general used-car dealerships.

Weaknesses:

Despite its simplicity, AutoManager lacks advanced features required by high-end sports car showrooms, such as detailed customization tracking and luxury-specific specifications. Its scalability is also limited, meaning it may struggle to support growing inventories or advanced performance requirements.

Furthermore, limited UI customization restricts the ability to deliver a premium digital experience expected by luxury customers.

How Our System Stands Out

The **Royal Sports Car Management System** is specifically designed to address the shortcomings of existing solutions by focusing on the unique needs of small to medium-sized luxury sports car showrooms.

Simplicity:

Unlike CDK Global and DealerSocket, which can be overly complex, our

system emphasizes a clean and intuitive interface. This reduces training time and technical dependency, enabling sales staff to focus on customer engagement rather than system navigation.

Cost-Effectiveness:

Developed using **C# and Microsoft SQL Server**, the system offers a more affordable alternative to enterprise-level solutions. It supports local hosting, eliminating mandatory recurring cloud subscription fees and reducing long-term operational costs.

Performance:

Performance optimization is a core design goal. The system supports real-time inventory updates, fast order processing, and streamlined test-drive bookings. This ensures consistent responsiveness, even as data volumes grow—an area where legacy systems often struggle.

Customization Without Complexity:

While maintaining simplicity, the system allows easy customization for luxury-specific attributes such as engine type, body style, and special modifications (e.g., *V8, Convertible*). This balance ensures flexibility without overwhelming users.

Overall, the Royal Sports Car Management System combines usability, affordability, and performance to deliver a solution tailored specifically for luxury automotive retail, avoiding the inefficiencies and high costs associated with generic, large-scale systems.

Data Gathering Approach

The development of this system was guided by a detailed research-based data gathering approach. An in-depth analysis of existing automotive management platforms—namely CDK Global [1], DealerSocket [2], and AutoManager [3]—was conducted to understand their core functionality, strengths, and limitations.

This comparative study highlighted common features such as inventory management, order processing, and reporting, while also revealing critical issues faced by smaller dealerships, including excessive complexity, limited customization, and high costs. Special attention was given to how these systems handle user interaction, data security, and alignment with customer needs.

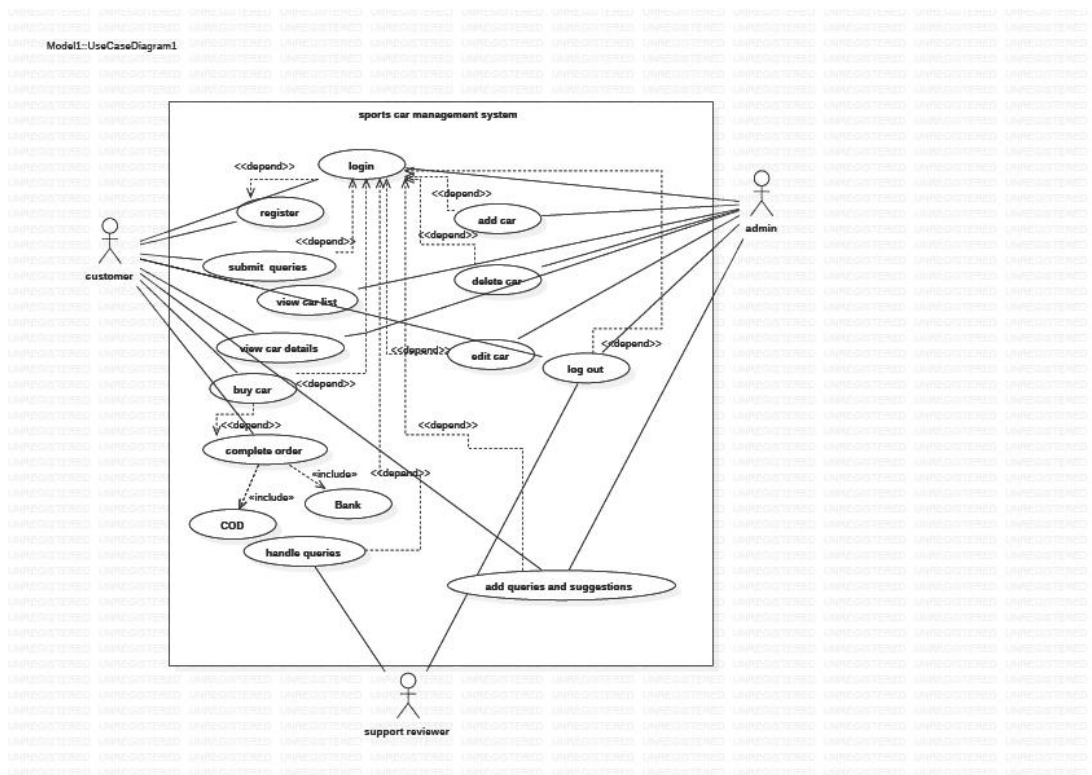
Rather than relying on formal surveys, insights were drawn from hands-on research, technical analysis, and personal experience as both a developer and automotive enthusiast. This practical perspective helped identify essential features such as role-based user management (Admin, Customer, Support), real-time pricing updates, and simplified purchasing workflows.

Additionally, operational challenges faced by sales staff were considered to ensure the system enhances daily workflows rather than complicating them. Features such as intuitive navigation, fast inventory updates, and streamlined “Buy Now” processes were prioritized.

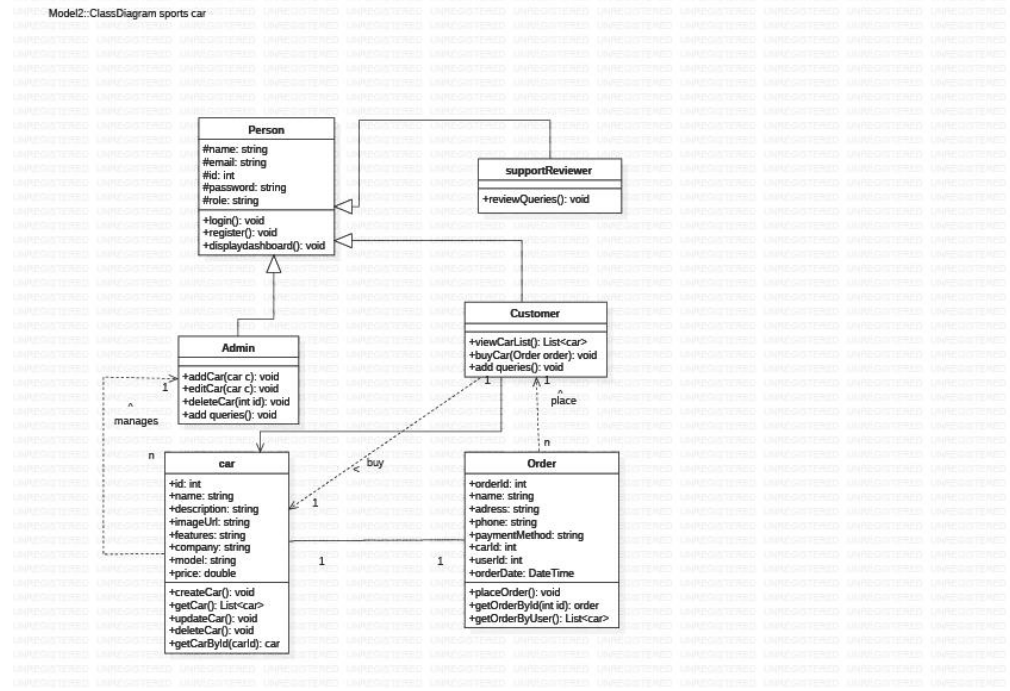
By combining industry research with practical observation, this project delivers a Sports Car Management System tailored to the needs of small luxury dealerships—where efficiency, simplicity, and customization are critical. The final system aims to reduce administrative overhead while enabling staff to manage inventory and serve customers effectively in a competitive digital marketplace.

UML Diagrams

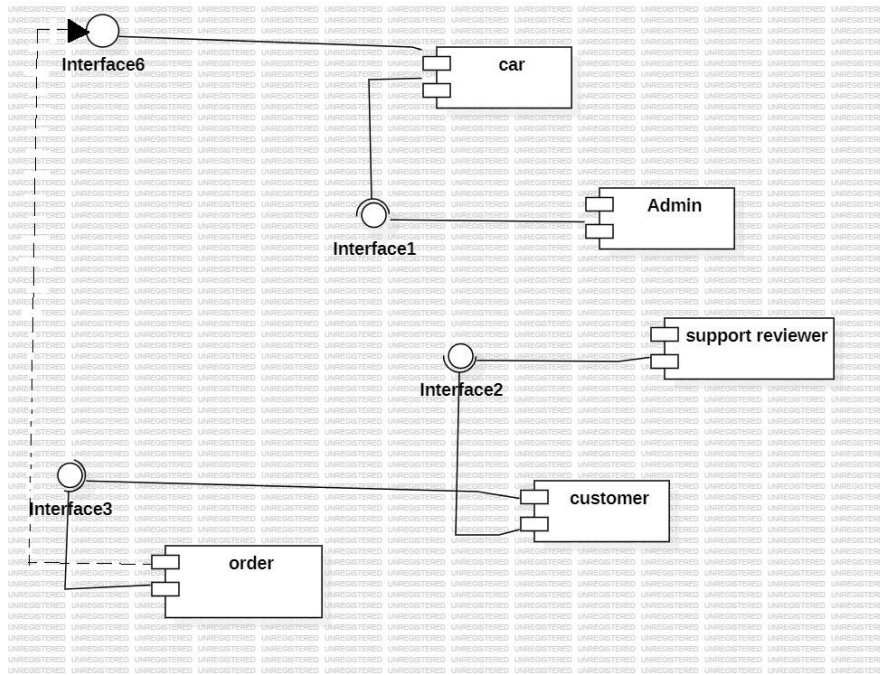
1. Use Case diagram



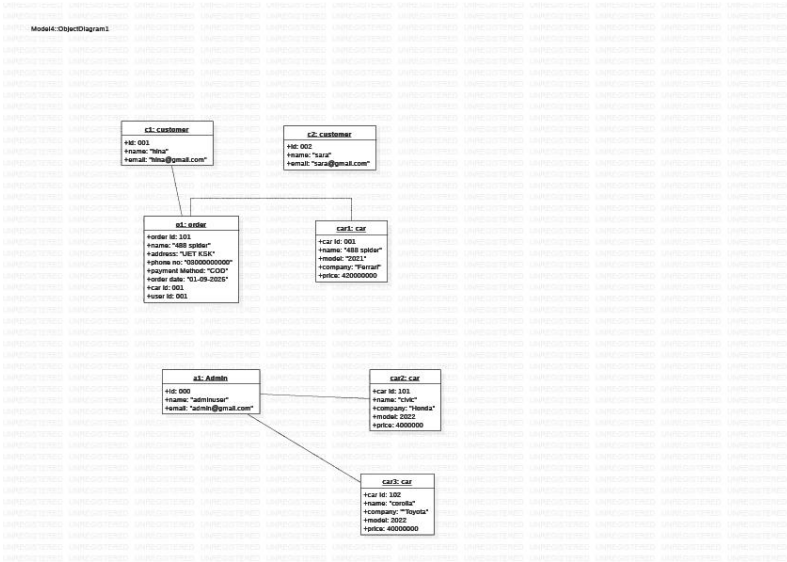
2. Class daigram



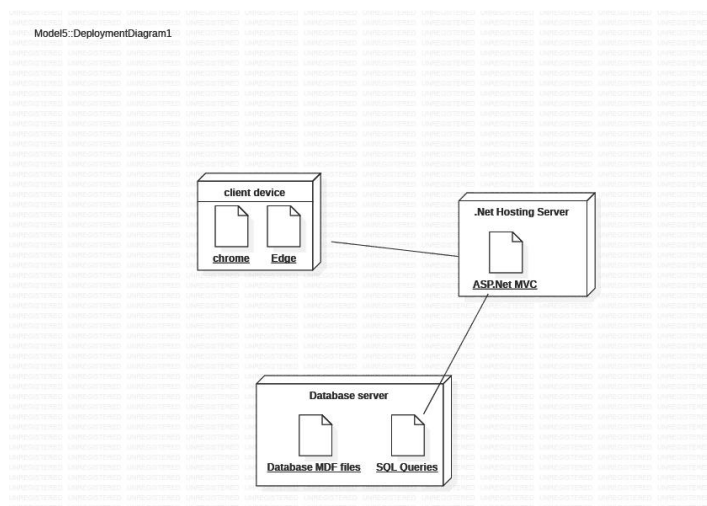
3. Component daigram



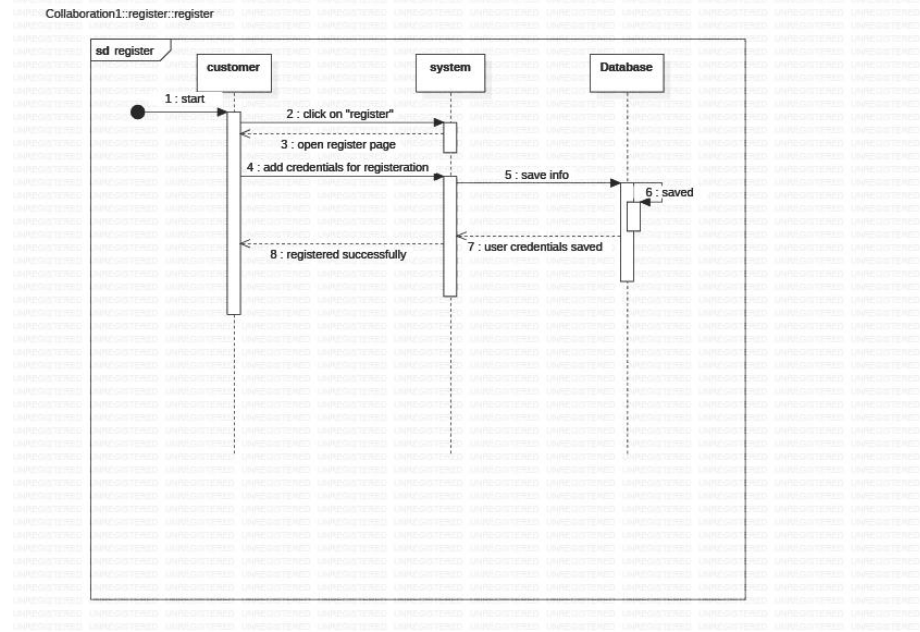
4. Object daigram

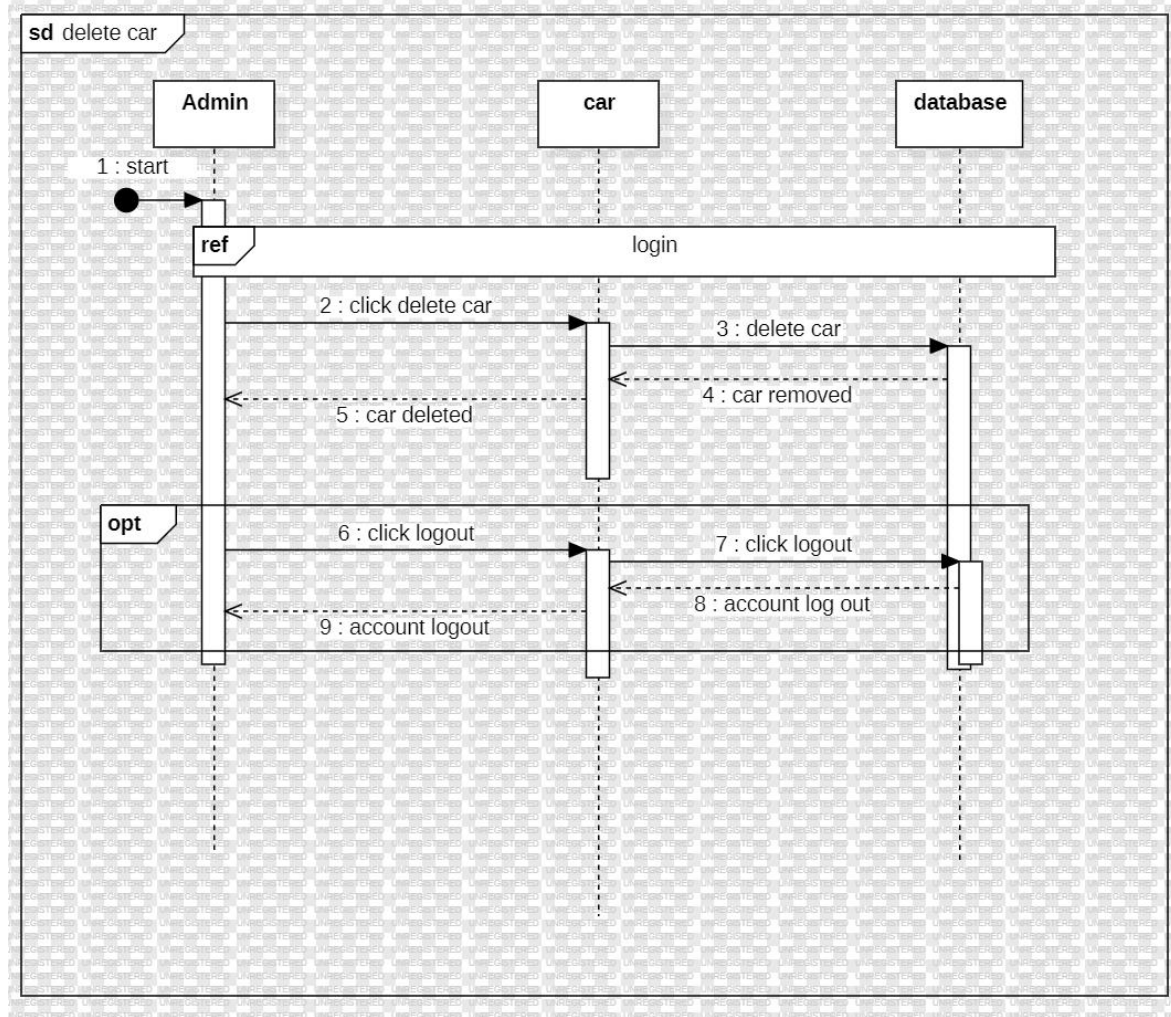


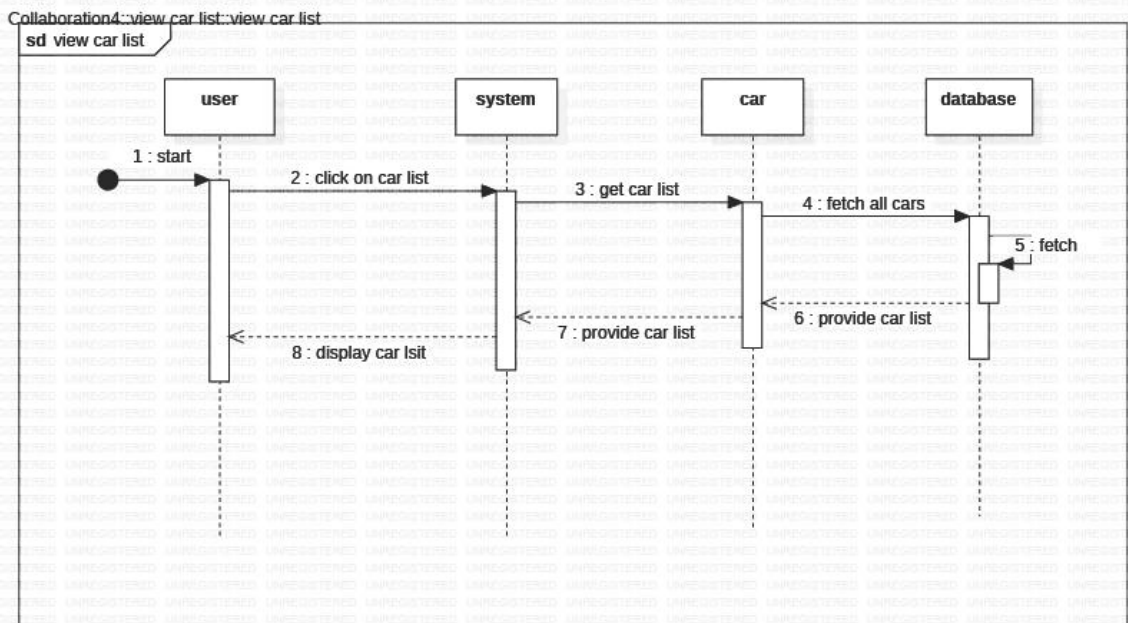
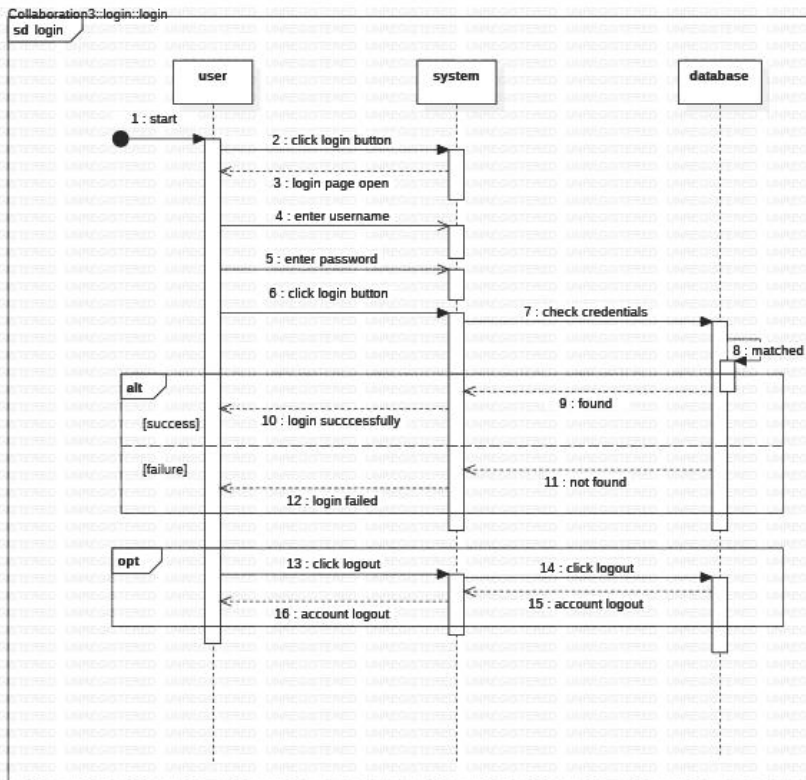
5. Deployment daigram



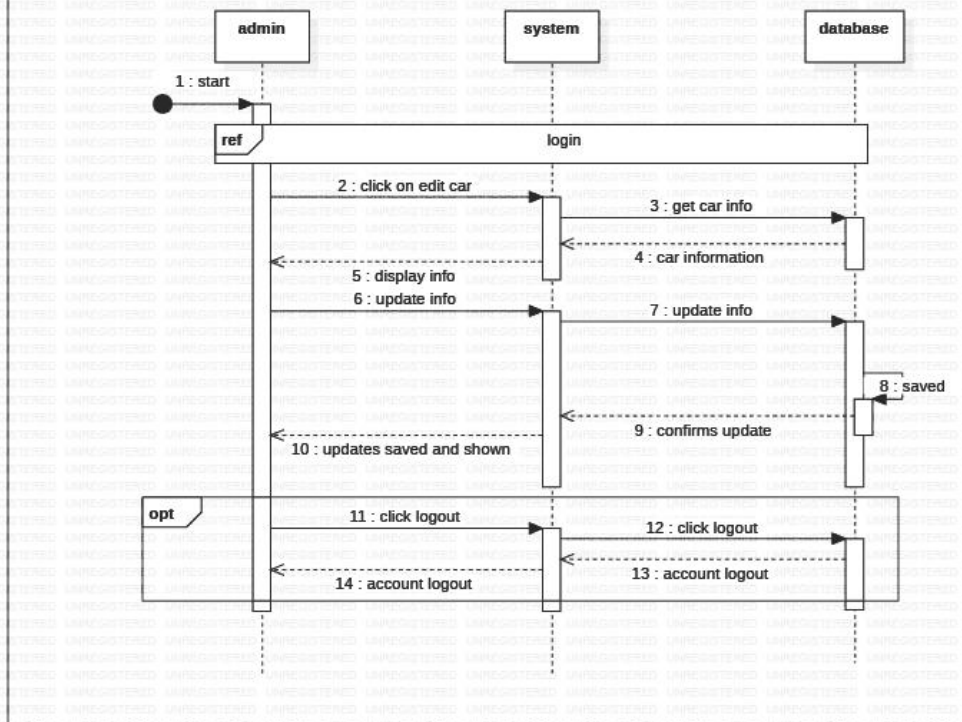
6. Sequence diagrams



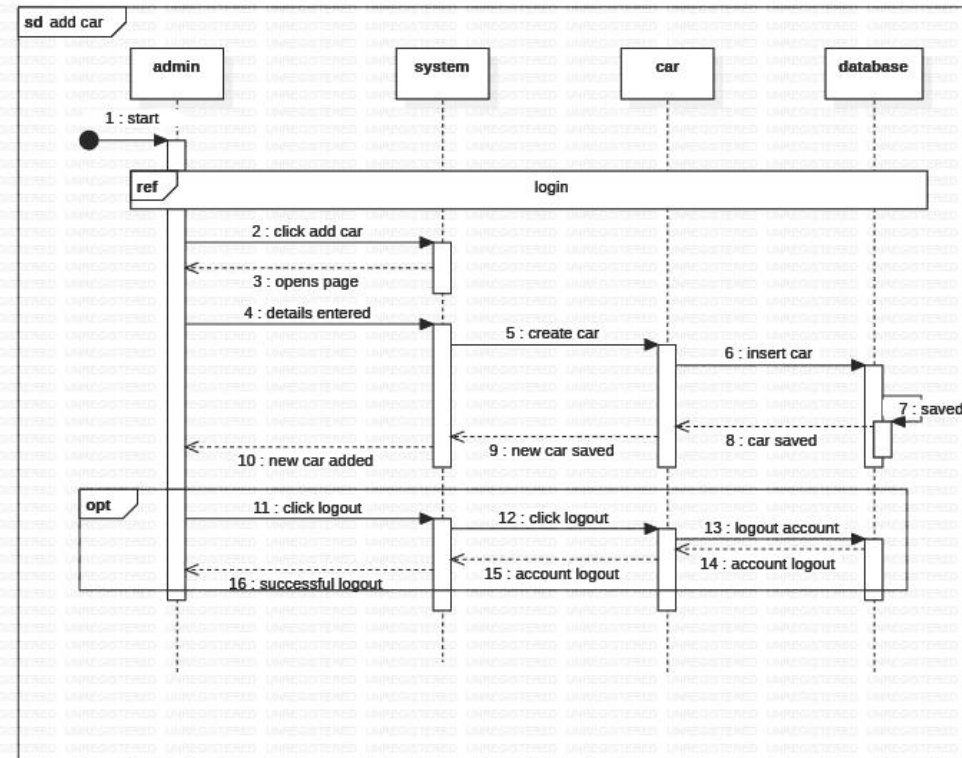




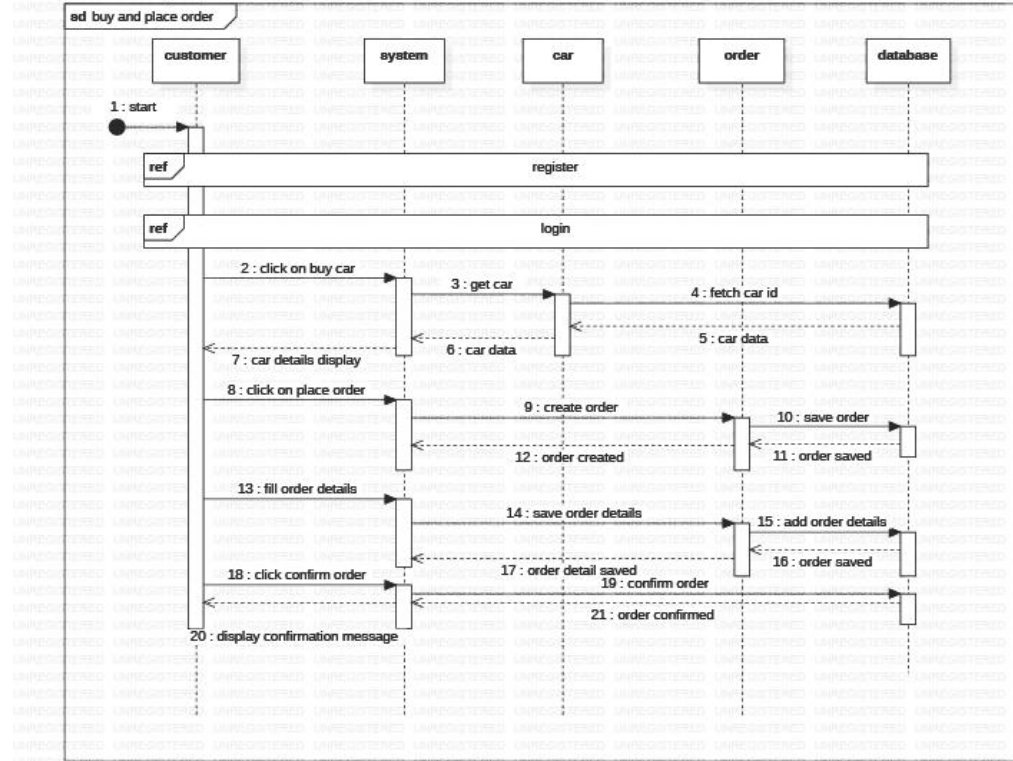
Collaboration5:edit car:edit car
sd edit car



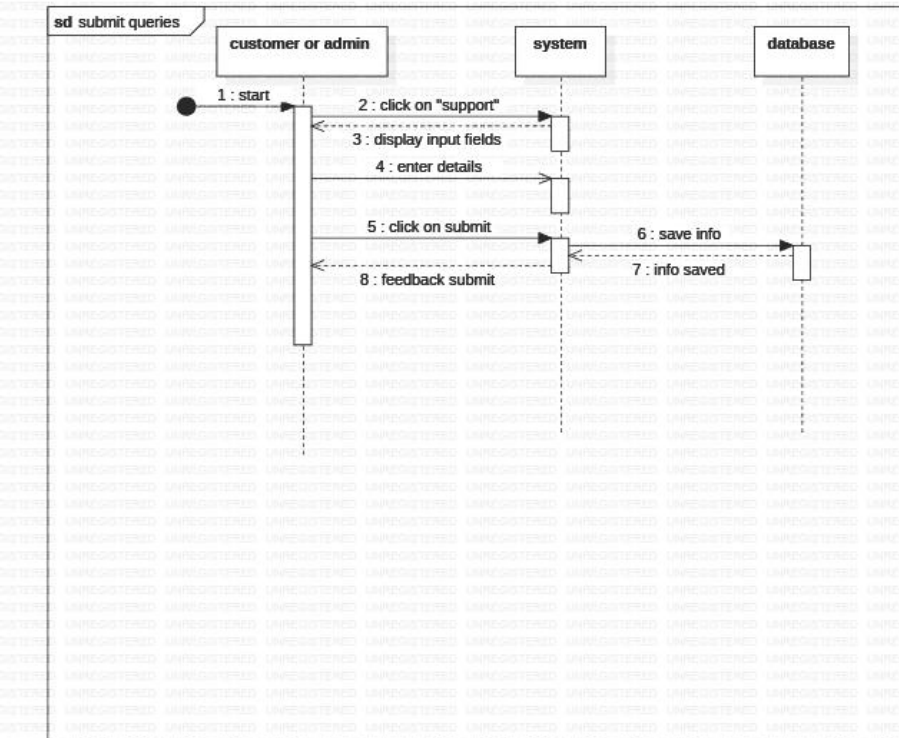
Collaboration6:add car:add car
sd add car



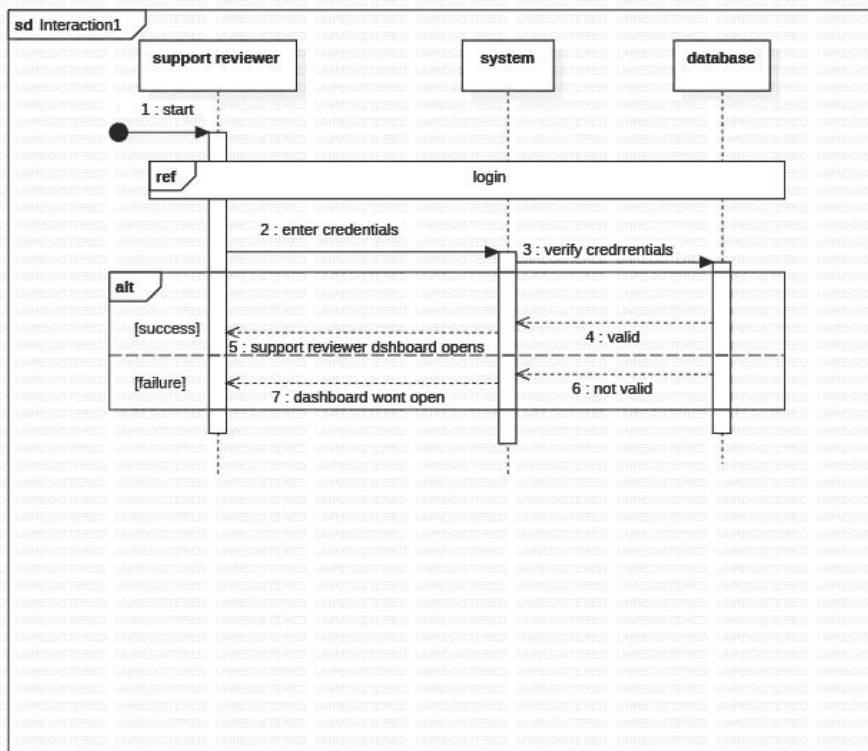
Collaboration7::buy and place order::buy and place order



Collaboration17::submit queries::submit queries

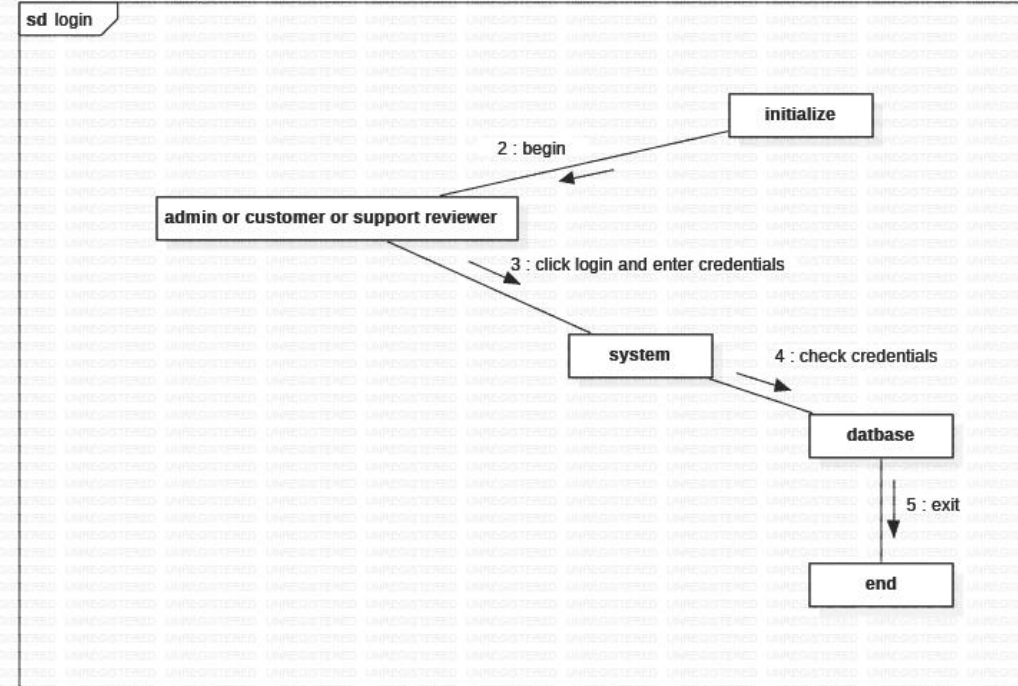


Collaboration18:Interaction1::view queries

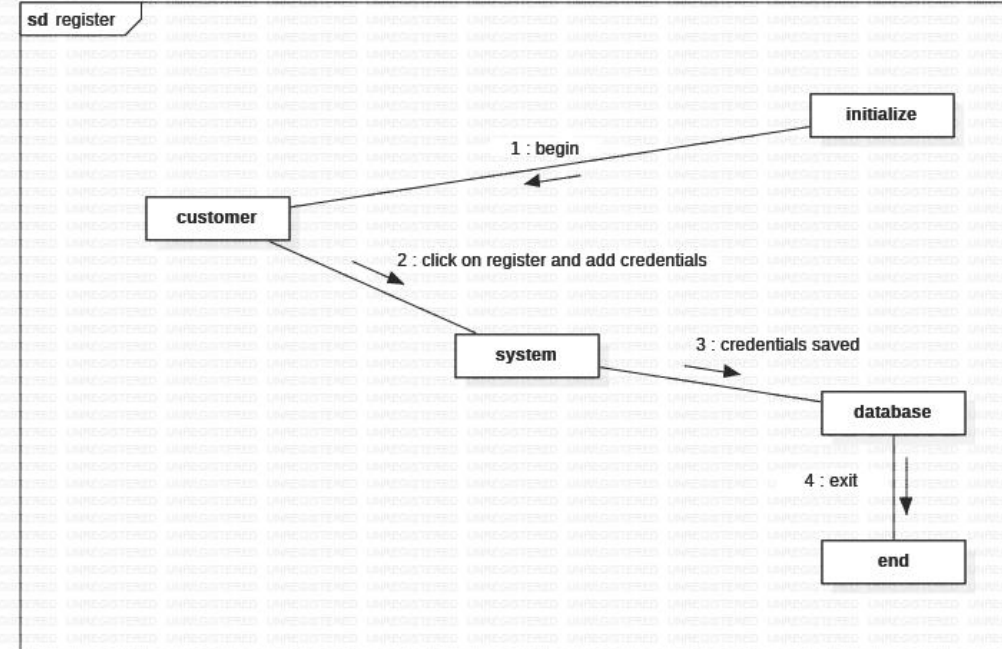


7. Collaboration daigram

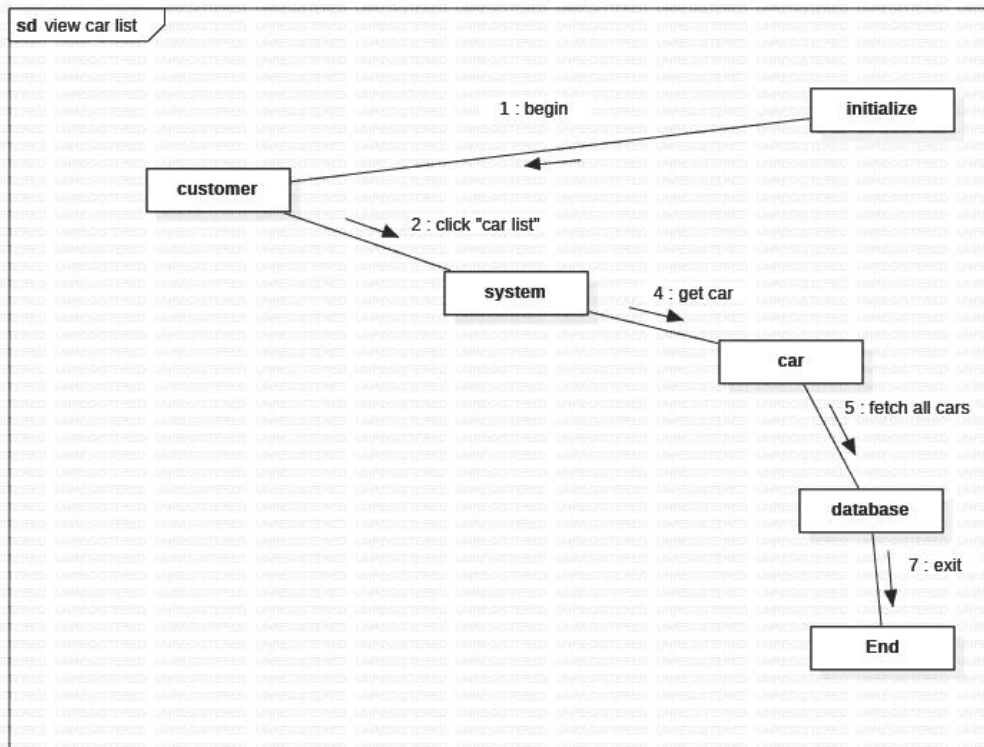
Collaboration8::login::login



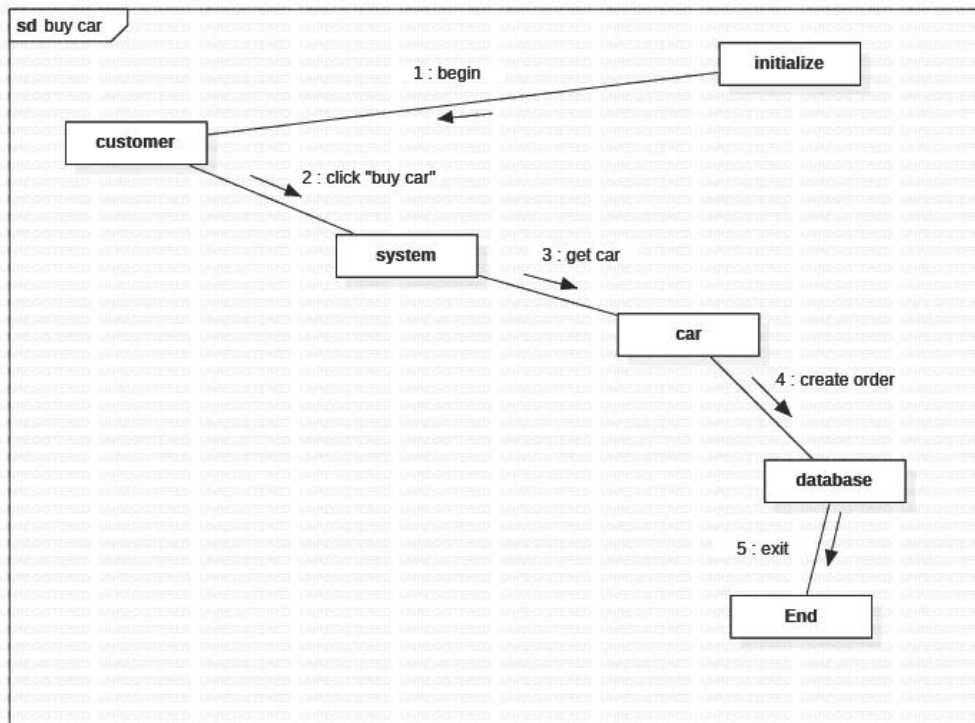
Collaboration9::register::register



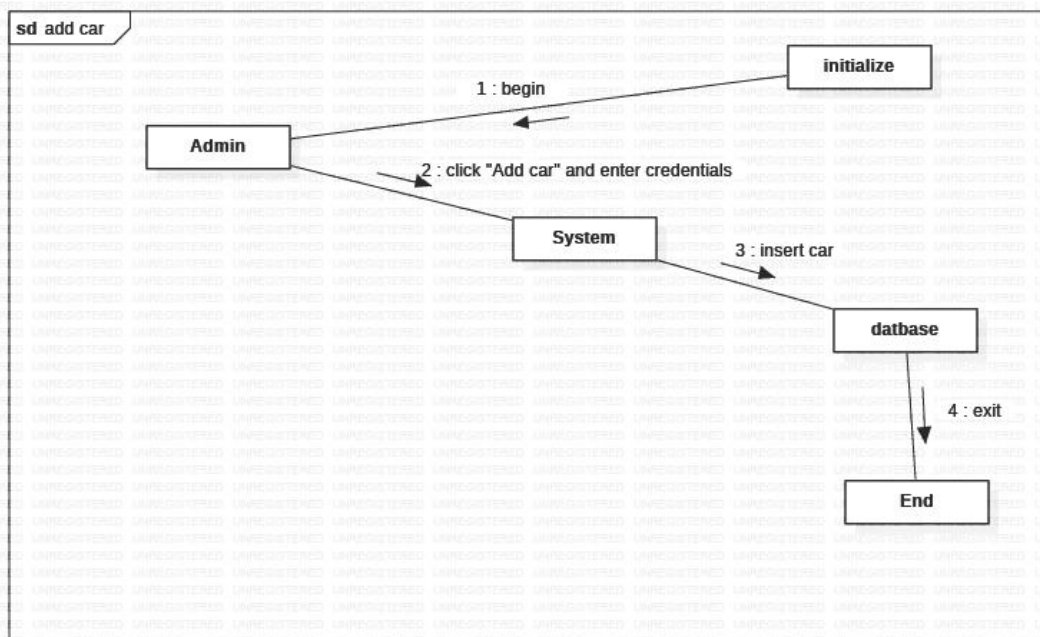
Collaboration10::view car list:view car list



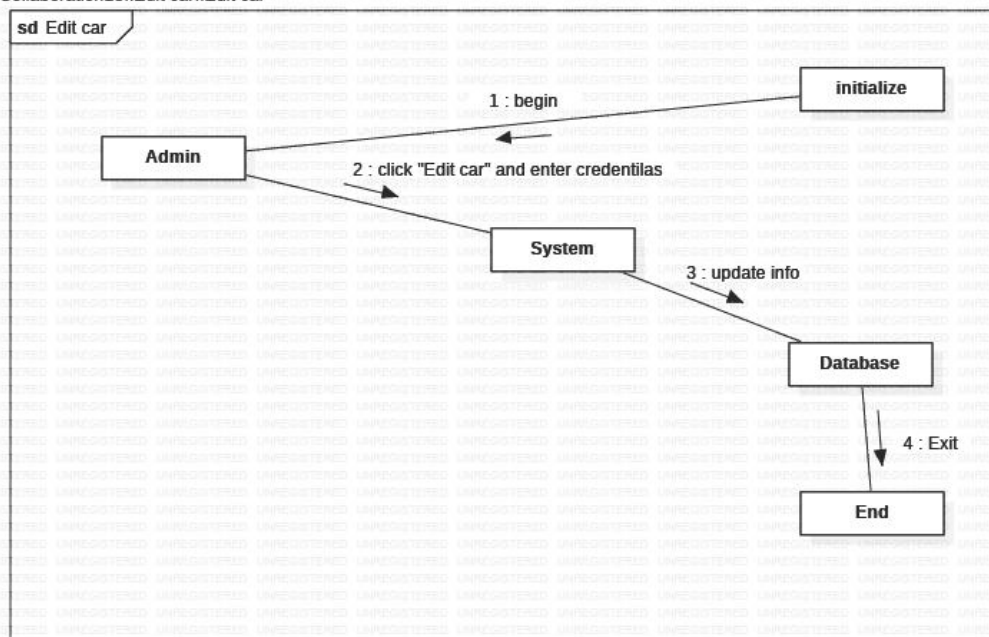
Collaboration11::buy car:buy car



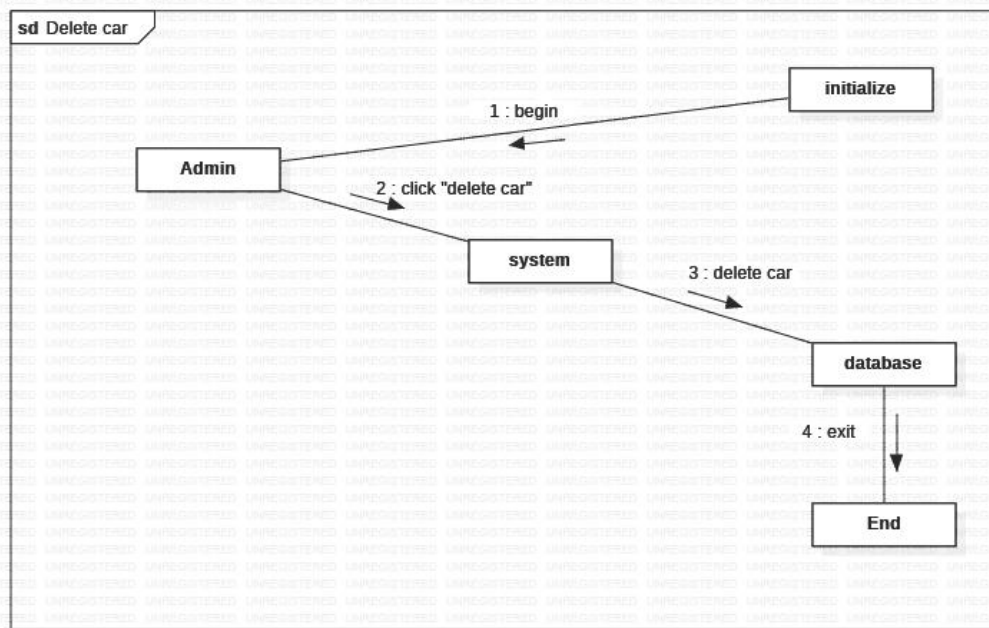
Collaboration12::add car::add car



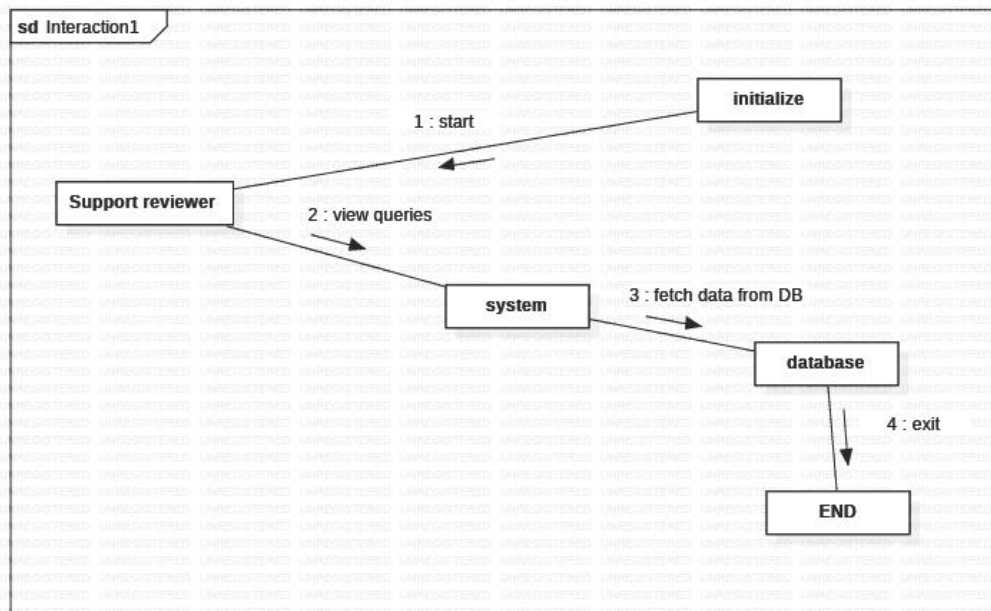
Collaboration13::Edit car::Edit car



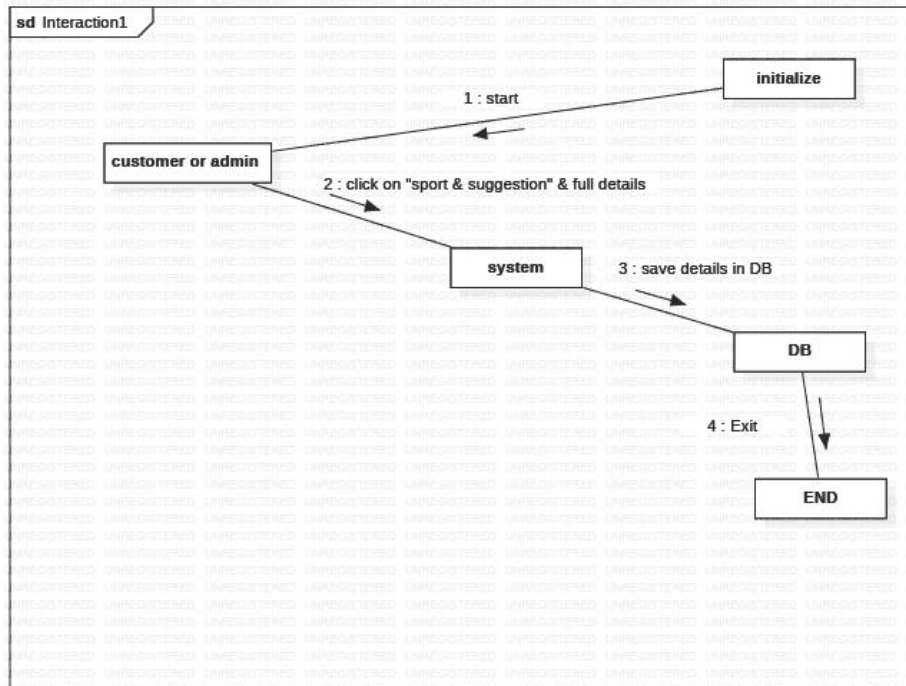
Collaboration14:Delete car::Delete car



Collaboration15:Interaction1::view queries

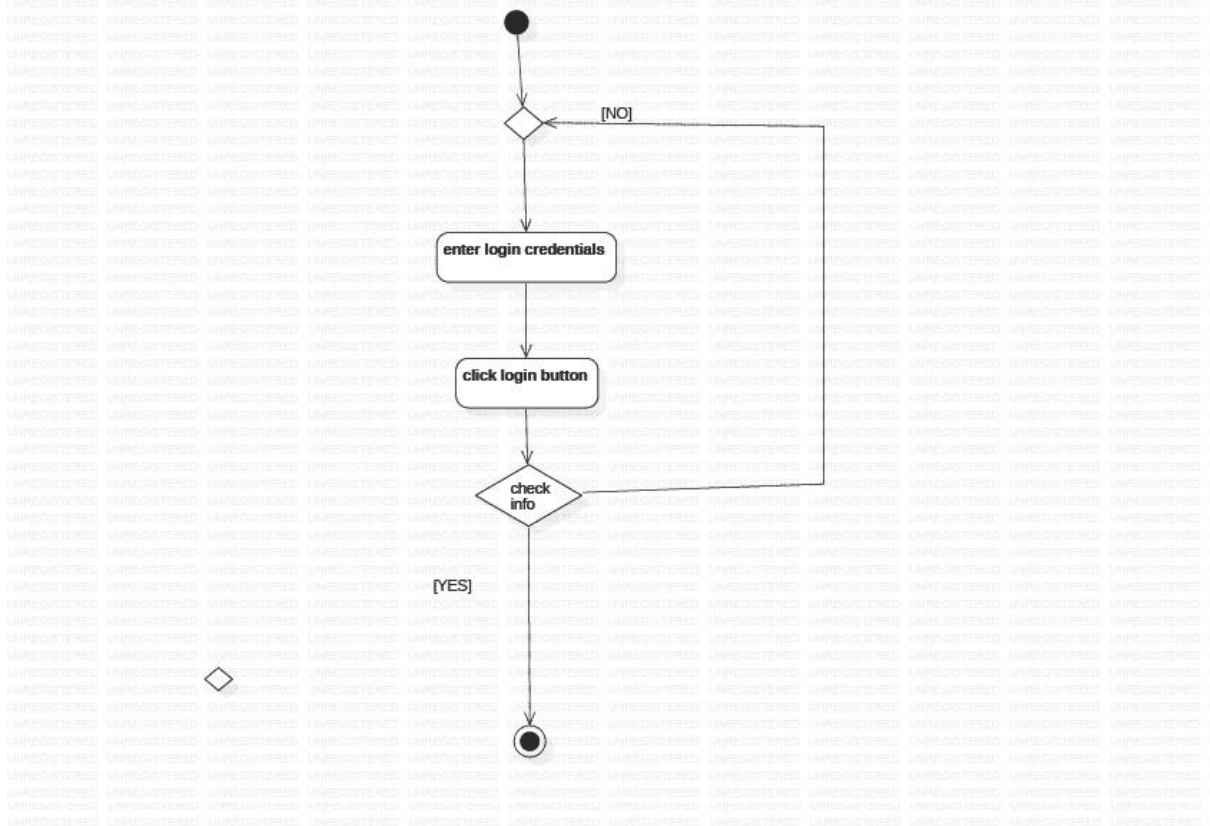


Collaboration16::Interaction1::submit query

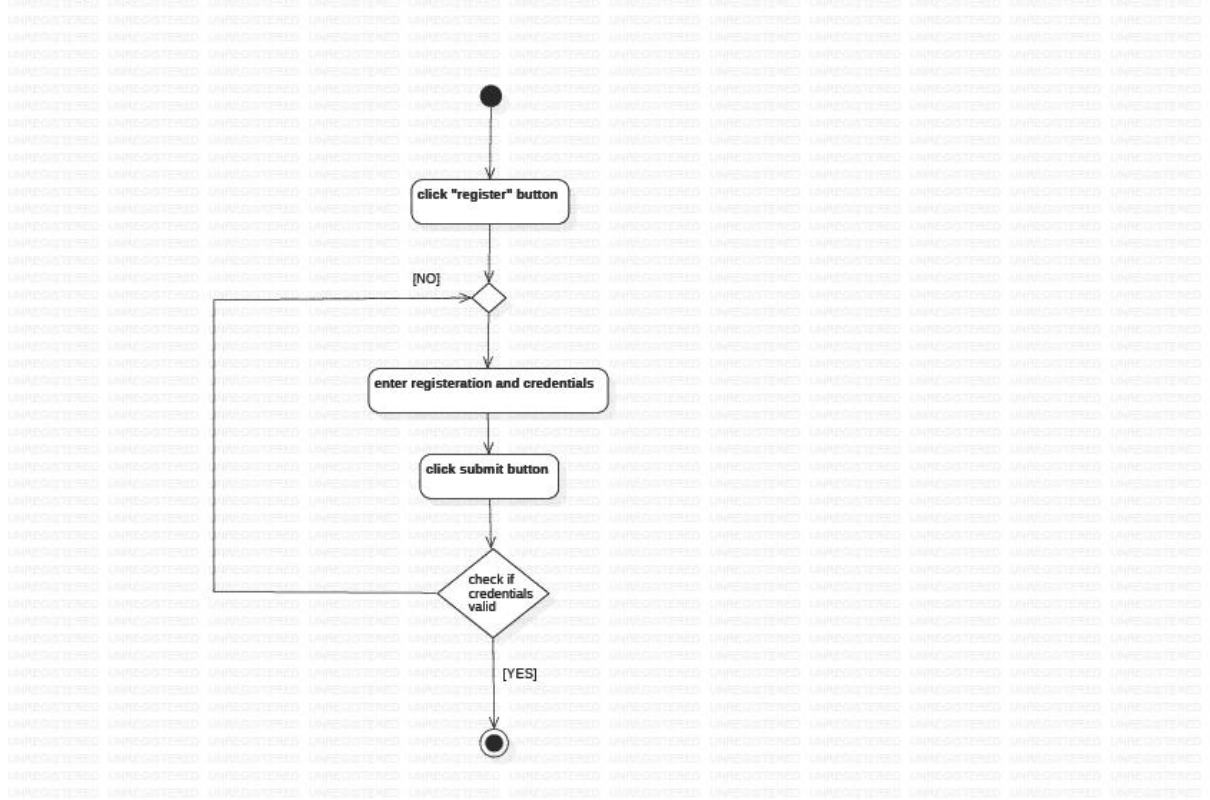


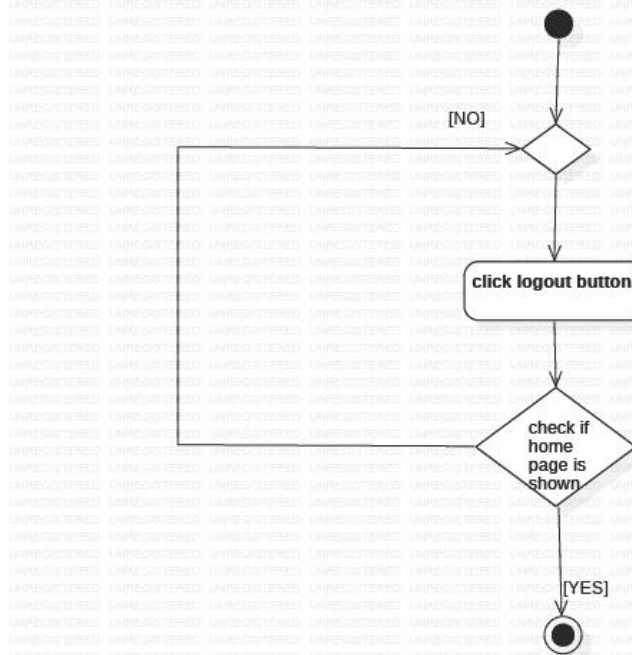
8. Activity daigram

Activity1:login

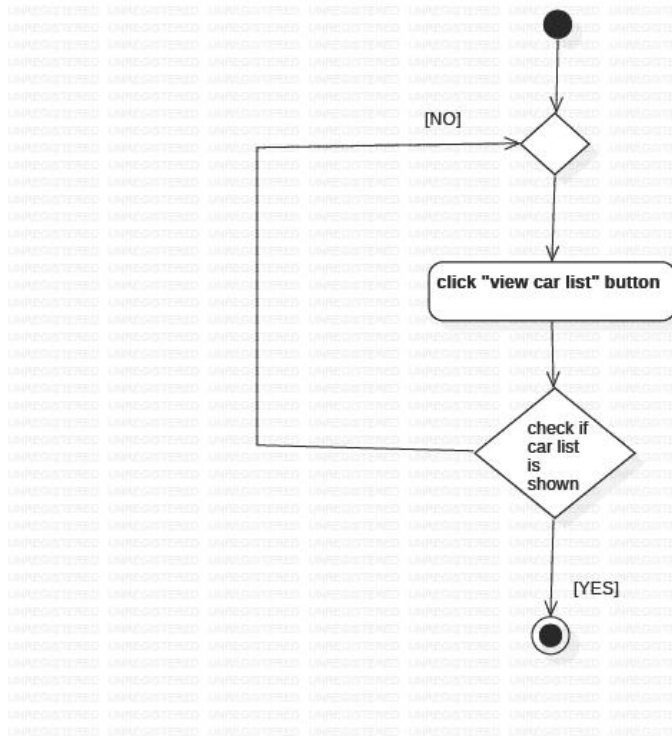


Activity1: Register

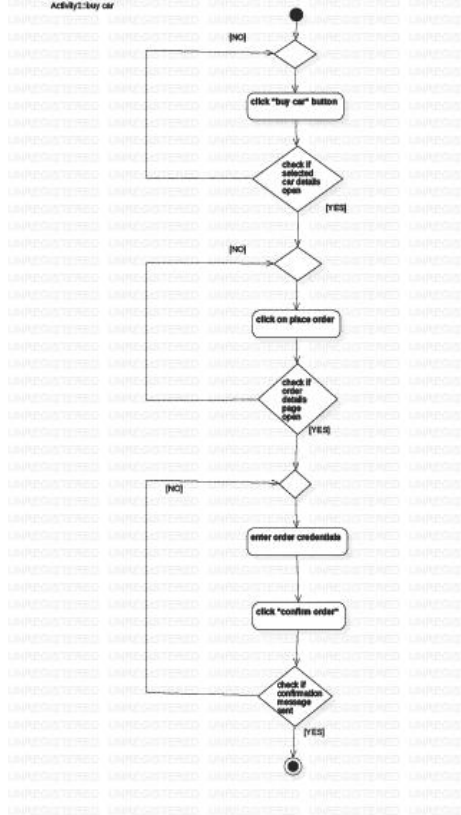


Activity1:logout

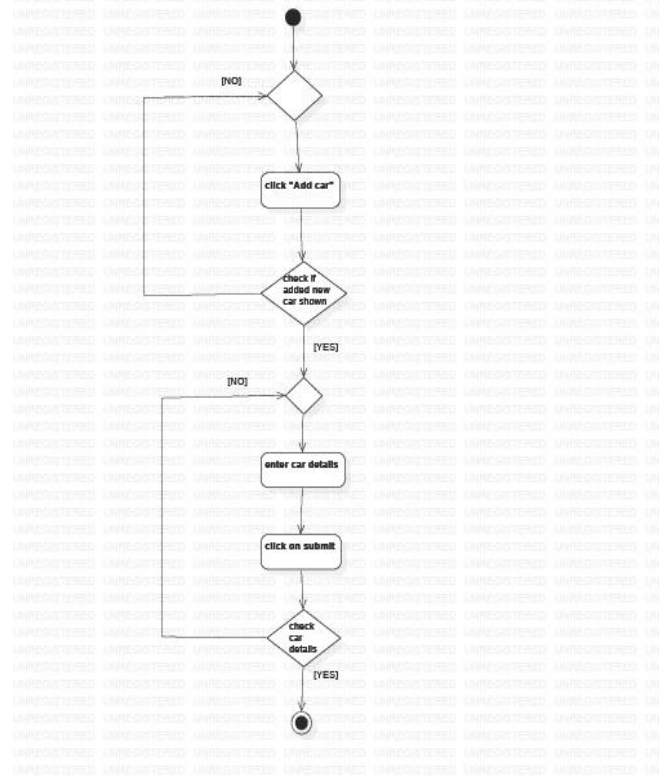
Activity1: view car list



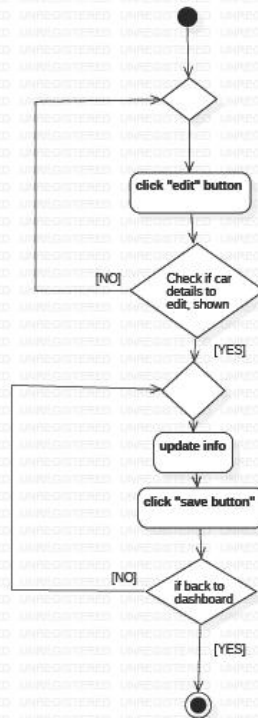
Activity1: buy car



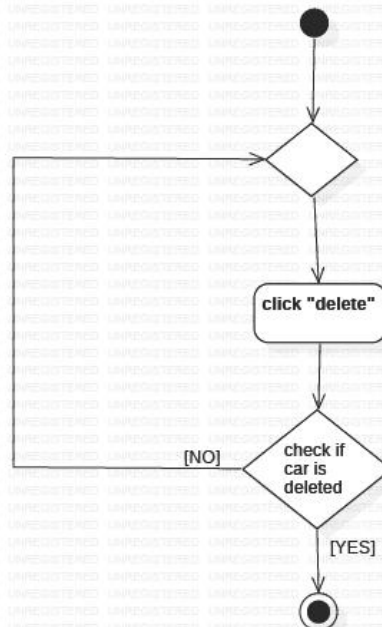
Activity1: add car

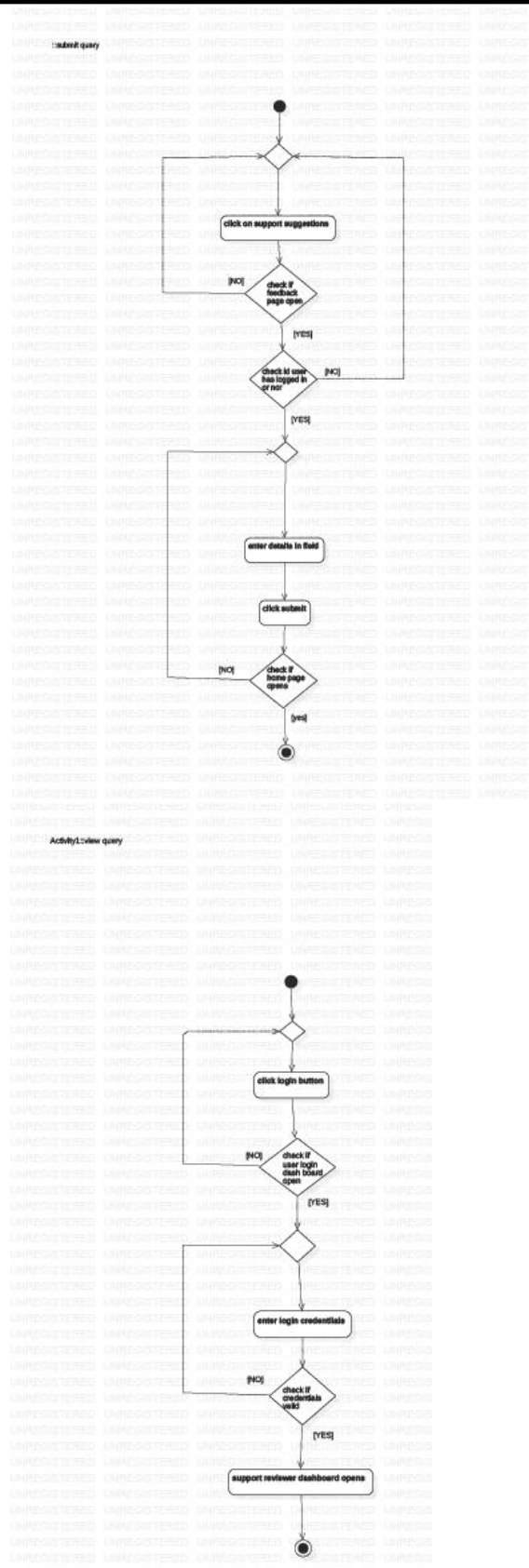


Activity1::edit car



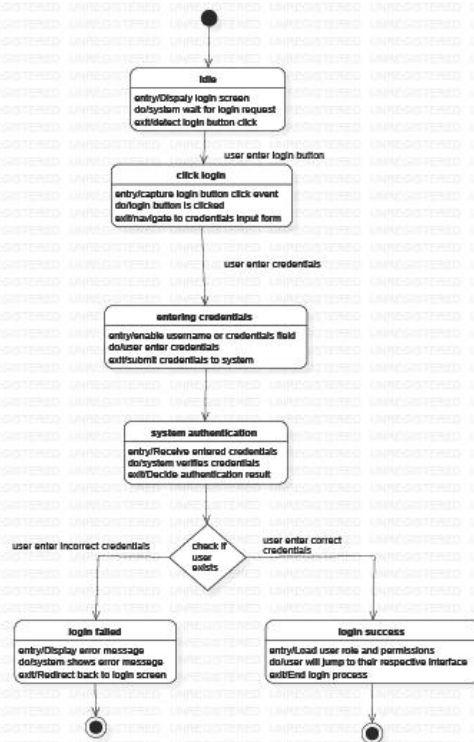
Activity1::delete car



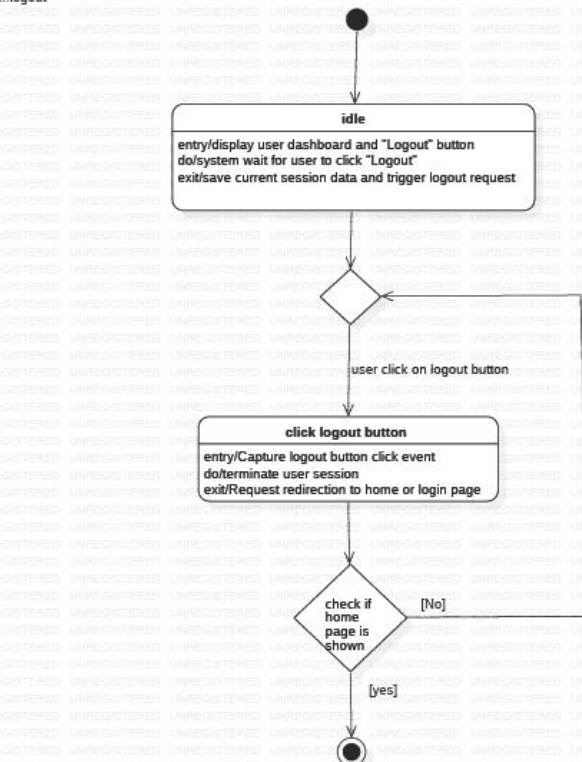


9. State daigram

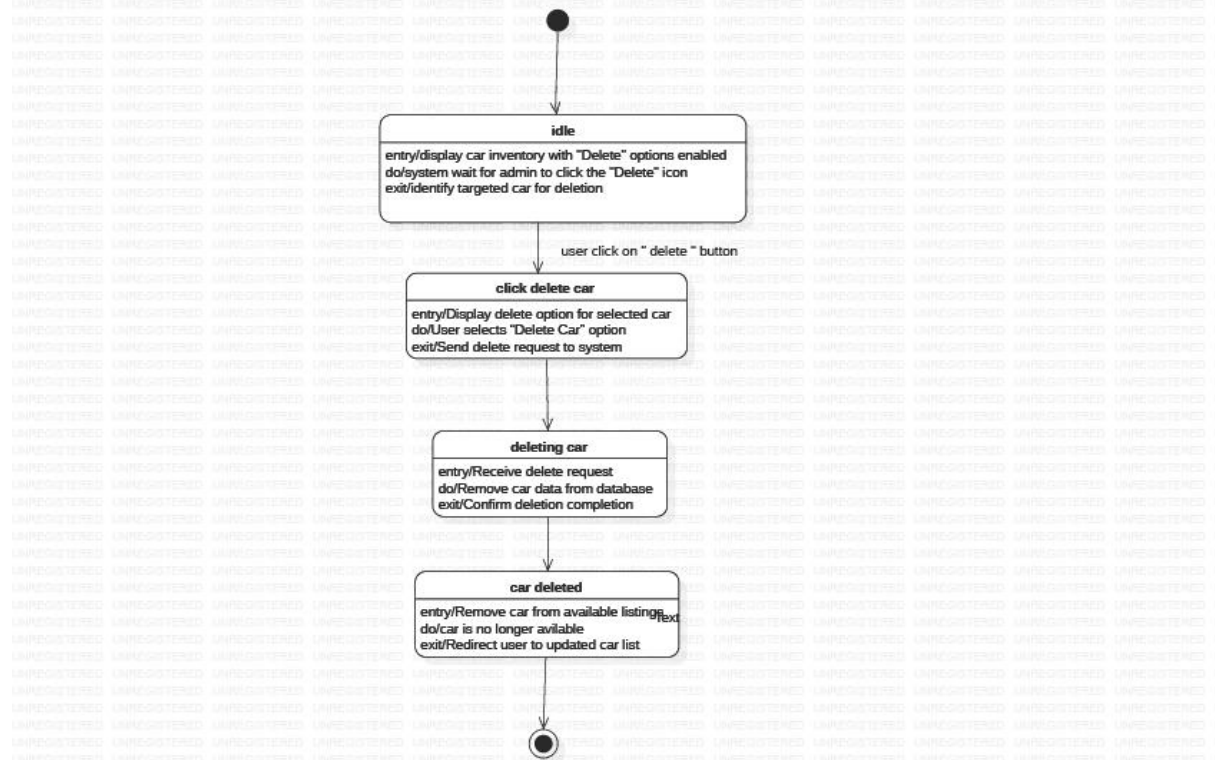
StateMachine1:login



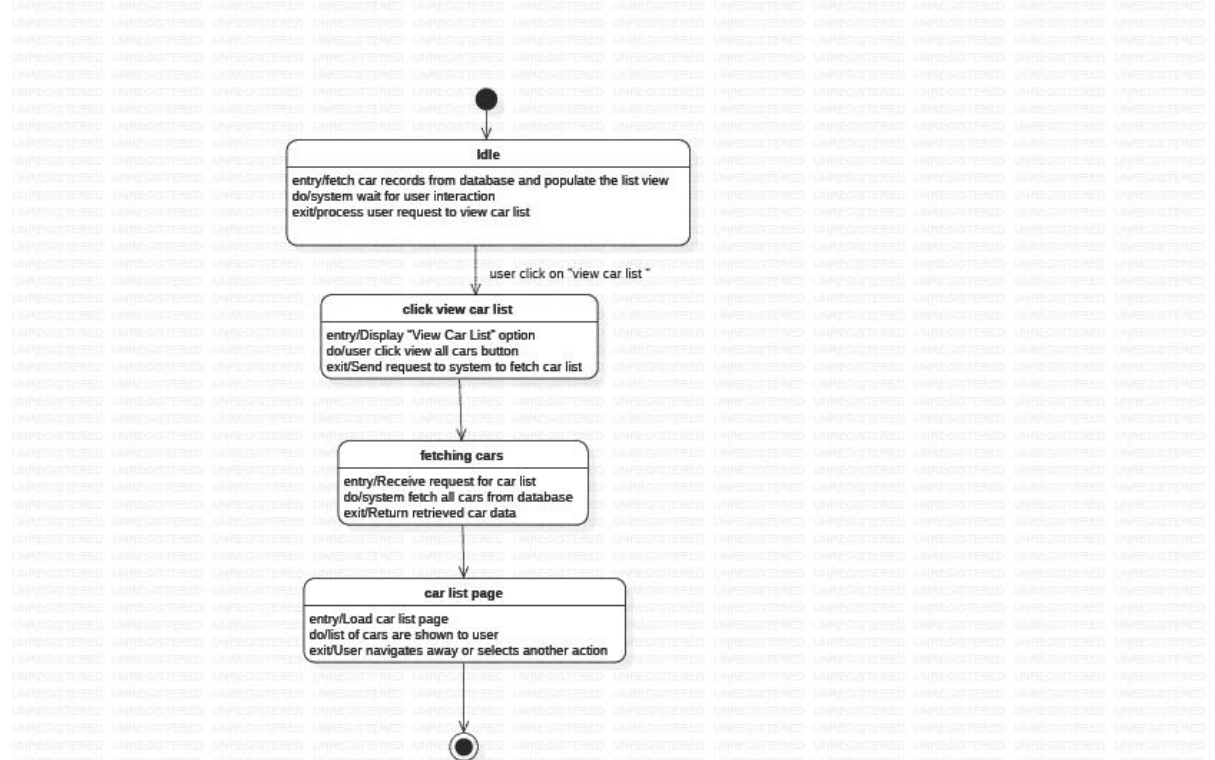
StateMachine2:logout

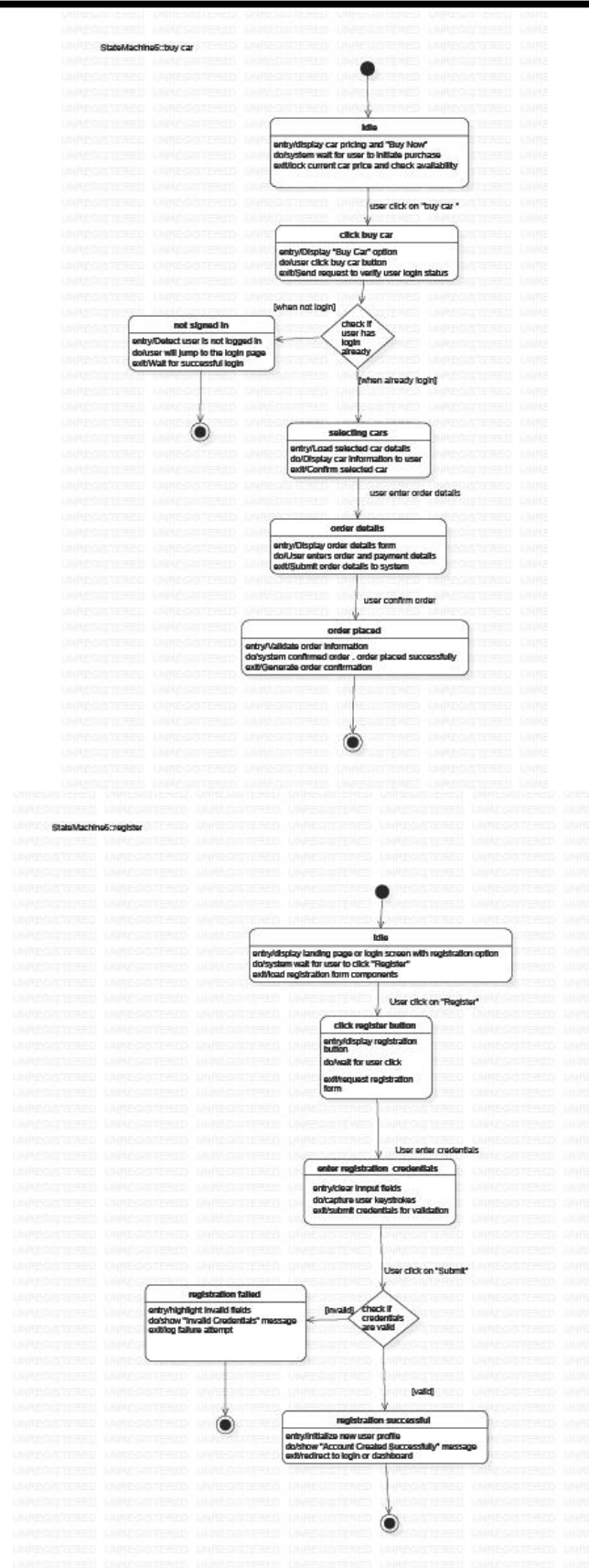


StateMachine3:delete car

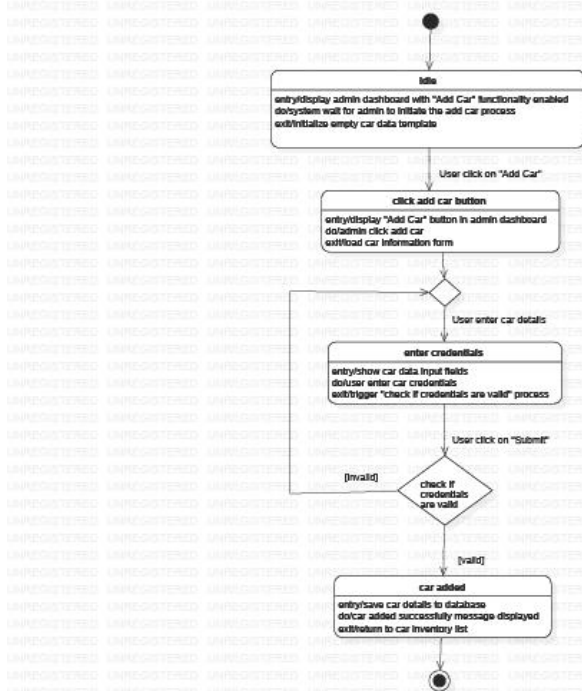


StateMachine4:view car list

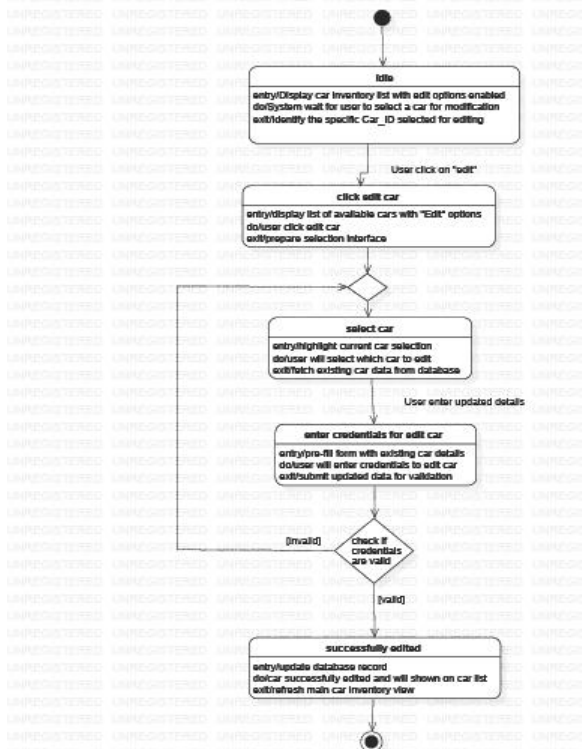




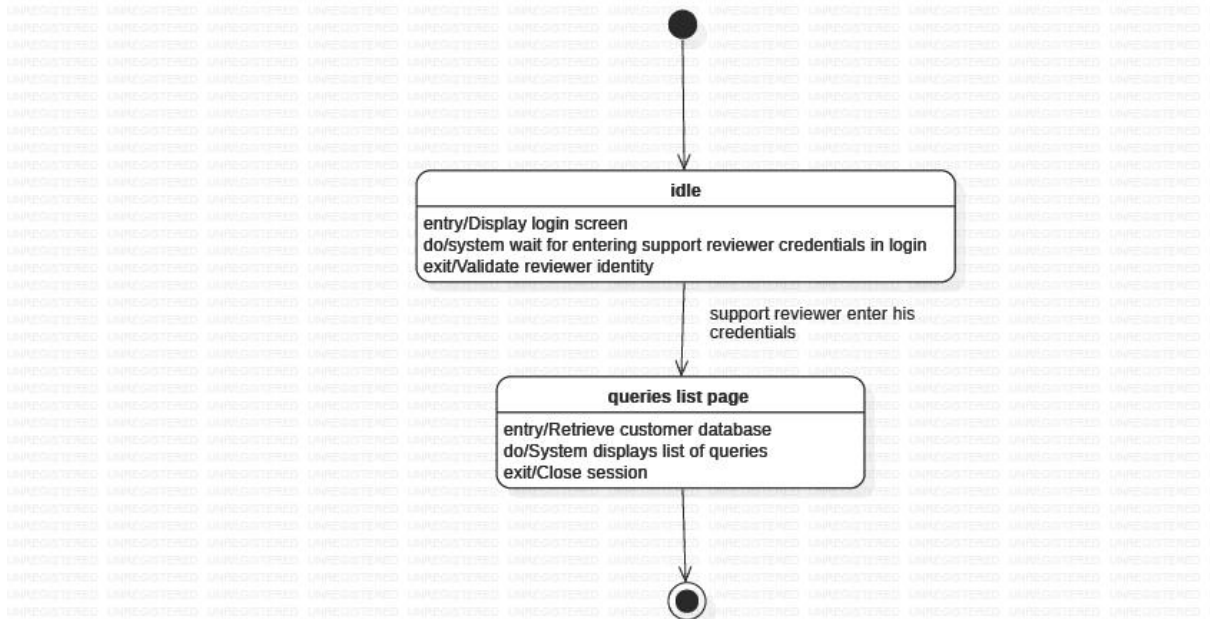
StateMachine7-add car



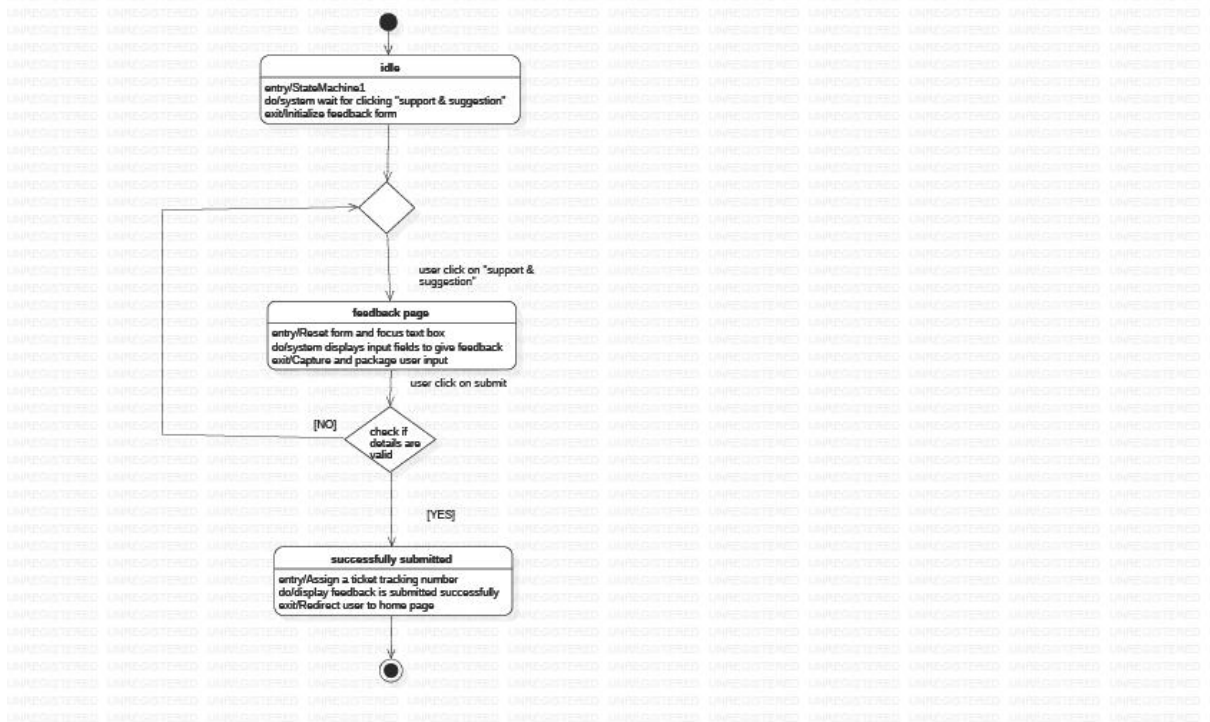
StateMachine8:edit car



StateMachine9::view queries



StateMachine10::submit queries



10. Package diagram

Model6::PackageDiagram1

